

Feeding of high-yielding bovines during transition phase

M. P. S. Bakshi¹, M. Wadhwa¹ and Harinder P. S. Makkar^{2*}

Address: ¹ Department of Animal Nutrition, Guru Angad Dev Veterinary and Animal Science University, Ludhiana-141004, India.

² Food and Agriculture Organization of the United Nations (FAO), Animal Production and Health Division, Rome, Italy.

***Correspondence:** HPS Makkar. Email: Harinder.Makkar@fao.org

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Abstract

Appropriate feeding during the pregnancy and post-pregnancy period determines both the milk production in the ensuing lactation as well as the reproductive efficiency of the animal. During this period, 3 weeks before and 3 weeks after parturition are very important because dramatic physiological and metabolic adaptations take place during this phase – called as the transition phase. Nutritional imbalances, caused by the low quality of desired nutritious rations, reduced appetite and low ingestion are responsible for depressed performance specifically during the transition period. The lack of knowledge and skill of farmers in feeding-management during this phase exacerbate the situation. The metabolic changes, metabolic adaptations and the nutritional-management strategies to ease out this phase are discussed in this review paper. With the advancement of gestation, the dry matter intake (DMI) of dairy cows declines and is at its lowest (1.7–2.0% of body weight) at parturition. The body condition score (BCS) parallels the DMI, but the aim should be not to lose more than one BCS (in the scale of 1–5) after calving. Simultaneously, with the growth of foetus the nutrient requirements increase and are at their peak before parturition as well as immediately after parturition. The milk production reaches its peak in about 5–8 weeks postpartum, while the diet consumption peaks at 10–14 weeks postpartum. Thus, dairy cows will typically suffer a 6–8-week period of negative energy balance during the postpartum period. The mammary energy requirements at 4-day postpartum are more than three times than that of uterus, with simultaneous increase in requirement of metabolizable protein, especially of methionine and lysine. The high incidence of metabolic and infectious diseases is responsible for the high incidence of inflammatory conditions, mostly immediately after calving. To prevent these diseases pro-inflammatory cytokine release should be avoided in postpartum. The mobilization of body reserves especially fat and protein; and hepatic gluconeogenesis takes place immediately after parturition leading to increased levels of β -hydroxy butyric acid and non-esterified fatty acids, which act as a gateway for a number of metabolic diseases. The transition phase can be eased by increasing DMI, feed conversion efficiency, density of ration, and protein supplements rich in rumen undegradable protein; by using anionic mineral mixture, optimizing roughage to concentrate (R:C) ratio, optimum physically effective fibre length of forages/silage/hay, dietary buffers, feed additives such as inflammation modulators, protected nutrients, glucogenic precursors and direct fed microbials/probiotics; and by following phased feeding programme.

Keywords: Bovines, Transition phase, Metabolic changes, Metabolic adaptations, Nutritional-management strategies, Inflammation modulators, Milk production

Review Methodology: More than 300 documents were screened and reviewed critically for writing this review. The documents included: Original research papers from Journals; Review articles; MSc and PhD theses; USDA and FAO documents; NRC and other feeding standards/ publications; ICAR publications; Feedpedia; Wikipedia; Surfing the internet: Google Scholar and Scopus; Proceedings of international conferences; Project reports; University extension bulletins; Experimental station bulletins; Technical bulletins; Annual reports of Ministry of Agriculture of different countries; Annual reports of Ministry of Animal Husbandry of different countries; Books; Book chapters.

Introduction

Milk yield of a dairy cow depends on four main factors: genetic potential, feeding regimen, herd management and health. As cows continue to improve genetically, the nutritional and management practices must also be improved simultaneously to allow the cow to produce to its potential. In developed countries with advanced livestock breeding programmes, the average herd size, though varies from country to country, can be considered 30–50 heads, yielding 18–40 litres milk/day [1]. On the contrary, most farmers in developing countries belong to the landless, small or medium land holding category: most rearing 5–15 adult cattle units yielding only 5–7 litres milk/animal/day [2] under non-intensive/low-input systems [3], exception being progressive dairy farmers, who rear crossbred cows yielding on an average 16 litres milk/animal/day [4]. Small and medium holder farmers are generally not aware of the impact of feed quality and balanced diet on the productivity of their bovines (mostly indigenous cows and buffaloes). Under field conditions, most farmers plan the feeding strategy only after the cow has calved rather than putting in place a well-planned feeding and management schedule for the last phase of gestation and ensuing lactation. It is mainly because of their lack of awareness and financial constraints.

Some developing countries are amongst the top milk producing nations. But in these countries, the high milk production is not due to high lactation yield, but mainly due to a high head count of low-yielding dairy cows. Approximately 37% of global lactating cows have a low feed nitrogen-use efficiency in milk (NUE-milk) of <10%. These cows account for only 10% of the milk production and 33% of the nitrogen (N) excretion from the global dairy sector. African countries such as Togo, The Gambia, Djibouti and Burundi have the lowest milk production and have NUE-milk of 3.3, 2.5, 4.1 and 4.1%, respectively. In contrast, ~30% of global lactating cows have NUE-milk between 21 and 25%, and these cows account for 53% of the milk production and 35% of N excretion globally. Countries such as the USA, Germany, UK and The Netherlands have the highest levels of milk production and have NUE-milk of 25.0–26.8% [5, 6]. These observations suggest that substantial improvement can be achieved in milk production and NUE-milk by providing all the nutrients in the appropriate quantity, quality and proportions. For getting the maximum output from a high yielder one must pay utmost attention to the feeding of a pregnant dry cow. The prenatal nutrition can influence postnatal growth, body condition, productive and reproductive performance in cattle. However, these effects are usually subtle and often ameliorated by postnatal nutritional experience [7]. Feeding during transition phase will determine both the milk production in the ensuing lactation as well as the reproductive efficiency of the animal.

Transition Phase

It refers to 60 days prior to and 60 days after calving; the most critical time within this period is the 14–21 days before and 14–21 days after calving [8–12]. This physiological stage is characterized by dramatic physiological and metabolic adaptations, and it is now well established that the cow is at the highest risk of succumbing to health disorder during this time [13]. From a rumen microbiota point of view, nutrition-induced changes during the transition period could play a vital role in providing optimal amounts of short-chain fatty acids (SCFAs), propionate in particular, to help alleviate the demands of the mammary gland for glucose and energy. In an effort to better regulate or attenuate the abrupt changes in nutrient supply and immune responses, one must look beyond the 6-week critical period in the transition phase. Changing to a shorter dry period, and reducing postpartum energy demands and fluxes in circulating metabolites may be a first step in offering practical solutions [14–16]. This coupled with the feeding of a ration formulated for a more favourable ratio between glucogenic and lipogenic precursors, may improve metabolism and overall health status [17].

Metabolic Changes

The major physiological changes that occur during this critical period of 3 weeks before and 3 weeks after the calving are: the decrease in dry matter intake (DMI) around parturition and changes in body condition score; sudden increase in nutrients' requirement that cows need for the growth of the foetus and onset of milk production [18, 19]; and metabolic adaptations required for meeting the nutrient requirements.

Decreased DMI

The DMI of dairy cows starts declining at 3 weeks prepartum, as during this period a foetus grows rapidly and various stresses can adversely impact hormone secretion [8, 19]. Average DMI for the pre-partum transition period has been reported to range between 1.7 and 2.0% of body weight (BW) [20]. However, this is not a constant value and varies with the breed and morpho-productive-type parameters such as quality of the ration fed, the stage of the transition period, body condition score (BCS) and parity. Grummer [8, 21] and Robinson and Garrett [22] indicated that the DMI declines during the last 3 weeks of gestation can be up to 30–35% (Fig. 1), especially in subtropical area such as Taiwan, where the summer temperatures usually reach 35°C during the day time. About 89% of that decline occurs at 5–7 days before calving [23]. The drastic decline in voluntary DMI is mainly because of the reduction in rumen volume due to the increased weight of foetus and uterus. Overcoming

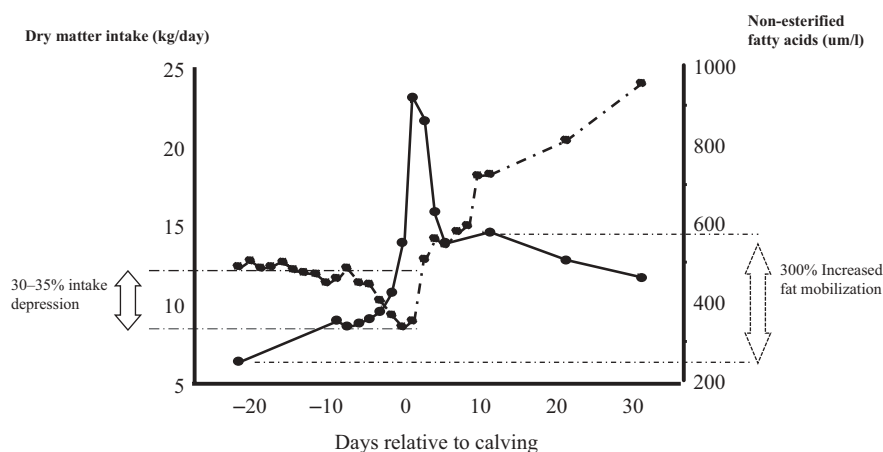


Figure 1 DMI and level of non-esterified FAs with the advancement of gestation (Adapted from Grummer [21]).

these limitations will require a significant shift in the dry cow feeding paradigm, with recognition that the postpartum disease conditions have their onset before calving.

Body condition score

The BCS during lactation cycle is probably the most important aspect of dairy cow management that facilitates a healthy transition from gestation to lactation. The negative energy balance (NEB) during late gestation and early lactation induces catabolism of body tissue to meet energy requirements, resulting in reduced BCS, rumen fermentation and milk production; and even worse, the possibility of triggering metabolic syndrome [24–27]. Dairy cows that are over conditioned (BCS > 4.0), have a much greater reduction in their feed intake immediately pre-calving, compared with cows with a lower BCS [28]. Mulligan *et al.* [28] indicated several benefits to health and reproduction when cows have a lower BCS at drying off and at calving, without any noticeable effect on milk production. While feeding such cows, the goal should be not to lose more than one BCS (in the scale of 1–5) after calving, which could be managed by maximizing DMI during the close-up period, which in turn depends on the dietary factors affecting hunger and satiety centres in the brain of mammals [29].

Increase in nutrient requirements

The dramatic increases in nutrient requirements following parturition have been quantified in terms of mammary demands for synthesis of milk lactose, fat and protein versus the lesser prepartum needs of the conceptus for glucose, fatty acids (FAs) and amino acids (AAs) [30]. Changes in calcium and other mineral requirements have been estimated similarly [31]. Bell [30] estimated that in a Holstein cow producing 30 kg milk at 4 days postpartum, the mammary requirements for glucose, FA and AAs are

respectively, 2.7, 4.5 and 2.0 times higher than those of the gravid uterus during late pregnancy. Similarly, additional nutrient requirements of non-mammary tissues, such as the liver and gut, also undergoing periparturient hypertrophy add to the challenge.

Energy requirements

The estimated mammary requirement for energy is three times higher than that of the uterus at 4-day postpartum [30]. Immediately after calving, at a time when feed intake has not reached its maximum, the demand of glucose for lactose synthesis increases to match the rapidly increasing milk production. The milk production reaches its peak in about 5–8 weeks postpartum, while the diet consumption peak is at 10–14 weeks postpartum. Thus, dairy cows will typically suffer a 6–8-week period of NEB during the postpartum period [32]. The propionate produced in the rumen during the fermentation of dietary carbohydrates is the major gluconeogenic precursor of glucose. Consequently, it is converted to glucose in the liver. In the last weeks of foetal development, the foetus uses an estimated 46% of maternal glucose taken up by the uterus [30]. Glucose demand in Holstein cows has been estimated at 1000–1100 g/day during the last 21 days of gestation, but it increases sharply after calving to approximately 2500 g/day at 21-day postpartum [33], confirming the earlier report of Bell [34]. Further, during the first week post-calving, there is a shortfall of 500 g glucose/day [33], which may be covered from body fat mobilization. The contribution of propionate to gluconeogenesis has been estimated to be 32–73% [35]. The muscle protein mobilization increased threefold during the first week after calving compared with prepartum values; fat mobilization is 300% (Fig. 1) [21]. Amino acids, lactate and glycerol are secondary substrates for gluconeogenesis in ruminants [36]. This poses an enormous challenge for the liver that has to synthesize all of this glucose from propionate and AAs. In addition, other tissues, organs and processes get adapted to reduced glucose availability [37]. Glucose is an

Table 1 Minerals and vitamins required for far-off dry cow and transition cow [42]

Macro minerals			Trace minerals			Vitamins		
	Far-off dry cow (% DM)	Transition cow (% DM)		Far-off dry cow (ppm DM)	Transition cow (ppm DM)		Far-off dry cow (IU)	Transition cow (IU)
Calcium	0.6	0.7	Cobalt	0.11	0.11	A	100 000	100 000
Phosphorus	0.26	0.30	Selenium	0.30	0.30	D	30 000	30 000
Magnesium	0.16	0.20	Zinc	60.0	60.0	E	1200	2000
Sulphur	0.20	0.20	Copper	12.0	18.0	Niacin (g)	6–12	6–12

Sodium, potassium and chloride requirements remain the same in far-off dry cows and transition cows, i.e. 0.10, 0.65 and 0.20%, respectively.

equally important energy source for the ovary, and its reduced availability in the beginning of lactation can negatively impact the re-establishment of ovarian activity after calving [38].

Protein requirements

Despite numerous studies on the level and quality of protein requirements before and soon after parturition on subsequent health and production, current recommendations are based mostly on predicted requirements rather than measured responses in terms of utilization of metabolizable protein (MP) and individual AAs [39]. Furthermore, feeding neither a glucogenic diet [40] nor supplementation of dietary protein [41] had a sparing effect on apparent gluconeogenesis from AAs in the early postpartum period. National Research Council [42] indicated that ~900 g MP is required per day during late pregnancy in Holstein cow and heifer. However, this estimation did not include an increment for synthesis of mammary tissue, which Bell *et al.* [39] approximated at ~120 g MP/day, resulting in an overall predicted requirement of between 1000 and 1100 g MP/day.

The predicted requirement cannot be met without supplemental protein sources and, in particular, of undegradable protein (UDP) sources. But excessive supply of dietary crude protein (CP) may be detrimental to performance [41, 43]. For example, Garcia-Bojalil *et al.* [44, 45] fed high-protein diets (~21% CP) containing either low (11.1%) or high (15.7%) concentrations of rumen degradable protein (RDP). They reported decreased milk yields and delays in both follicular development and luteal function in cows fed the ration containing 15.7% RDP. These effects may be attributed to the apparently impaired capacity of the liver to detoxify ammonia to urea [46, 47]. Feeding of excessive RDP results in elevated blood urea nitrogen (BUN) level, or both and elevated milk urea nitrogen (MUN); affecting the uterine pH in heifers, which has been associated with reduced fertility [48]. Increased BUN or MUN concentrations are highly negatively correlated with fertility in cows [48–50]. The excess ammonia and elevated pH levels in the rumen causes Ca^{2+} and Mg^{2+} to form insoluble complexes with phosphorus making them unavailable for absorption encouraging both hypocalcaemia and hypomagnesaemia, which causes parturient paresis immediately after parturition in cows.

Requirements for metabolizable methionine (Met) and lysine (Lys) should also be considered during this period of need, as these are the AAs that determine the yield of milk/milk protein. In comparison with corn, barley has more protein, Met, cysteine, Lys and tryptophan. This highlights the potential contribution of barley to meeting the protein requirements of high-producing ruminants. Therefore, quality protein supplements should be selected to meet the requirements.

Mineral requirements

Minerals and vitamins are essential nutrients and are required for mammary gland development, growth of the developing calf, and support of the immune function. Nutritional programmes that deliver the optimal amounts of minerals and vitamins can promote good health and productivity. The mammary requirement for calcium to produce 10 kg of colostrum on the day of parturition is more than double than that for foetal growth in late gestation [31]. The onset of lactation places such a large demand on mechanisms of calcium homeostasis that most cows develop some degree of hypocalcaemia at calving. In some cases, concentrations of plasma calcium become too low to support nerve and muscle function, resulting in parturient paresis or milk fever [51]. Adaptations to increase the blood supply of calcium very soon after calving include increased intestinal active transport, increased resorption of bone stores and decreased urinary excretion of calcium [52]. Besides Ca, other minerals and vitamins are associated with a variety of metabolic conditions (Table 1).

Infectious diseases and metabolic disorders

Nutrient deficiencies and excesses can be responsible for metabolic and infectious diseases, and thus for animal welfare impairment and overall may result in poor efficiency and income for the farm. The high occurrence of metabolic and infectious diseases is responsible for the high incidence of inflammatory conditions [53] in the transition period (mostly after calving). In high-yielding cows in particular, this occurs more often during the transition period and the inadequate increase of DMI can be an important causal factor. Retained placenta, metritis and endometritis are

diseases of immune function in the transition period, starting at least 2 weeks before calving [54, 55]. Nevertheless, low DMI can also be a consequence of health disorders that induce inflammatory conditions (such as metritis, mastitis, lameness, etc.) through the release of pro-inflammatory cytokines, i.e. proteins (~5–20 kDa), which are involved in autocrine signalling, paracrine signalling and endocrine signalling as immune-modulating agents. Cows with the worst inflammatory indices appear affected by a NEB and increased levels of β -hydroxy butyric acid (BHBA). Some surveys on the transition period of dairy cows showed a relatively high frequency of inflammations, mostly sub-clinical, which besides infections could be attributed to: tissue damages, oxidative stress, digestive upsets, heat stress, placenta–uterus interactions, dystocia, excess of energy supply, among others. To avoid inflammatory phenomena to the greatest extent possible, it is recommended to take advantage of the dry period to prevent infectious and metabolic diseases, as well as any other cause of pro-inflammatory cytokine release in the postpartum [56].

The pattern of production diseases gives an indication as to which area of nutrition requires attention during the transition phase. For example, a high incidence of lameness is more indicative of rumen acidosis, whilst a high rate of displaced abomasum could be due to subclinical ketosis (SCK) and/or hypocalcaemia. It has been proposed that the key areas to monitor are NEB, BCS, rumen health, calcium status, trace elements and antioxidant status in blood.

The shortfall in glucose availability after parturition is one key driver of ketosis, a potentially deadly metabolic disorder [57]. Milk fever, ketosis, retained foetal membranes, metritis and displaced abomasum primarily affect cows within the first 2 weeks of lactation [18]. Physical and metabolic stresses of pregnancy, calving and lactation contribute to the decrease in host resistance during the periparturient period [58]. A number of studies revealed that higher levels of non-esterified fatty acids (NEFAs) and BHBA caused immune suppression around calving [59, 60]. During 2 weeks before and after parturition the T-cells population exhibit a significant decline, which contribute to the immune-suppression in dairy cows at calving [61], this leads to increased susceptibility to mastitis and other infectious diseases, which should not be overlooked, rather the animals should be treated immediately [58]. Other diseases that are not clinically apparent during the first 2 weeks of lactation (laminitis, ovarian cysts, endometritis) can be traced back to early lactation [53]. Imbalanced feeding in the last phase of gestation has been reported to effect birth weight of new born calf, postpartum oestrus and obviously productive performance of cows [62, 63]. Critical physiological events that have to be targeted during the transition period include adaptation of the rumen to the higher energy diet that will be fed in early lactation, maintenance of normal blood calcium concentration, a strong immune system throughout the peripartum period,

and maintenance of a slightly positive energy balance up to the time of calving [53, 64].

Endocrine regulation of gluconeogenesis, ketogenesis and lipid metabolism includes insulin, glucagon, somatotropin, catecholamines, cortisol, thyroid hormones and leptin [33, 36]. Glucose levels in a prepartum dairy cow are high [65]. As a result, insulin concentrations are higher before calving than after calving and glucagon has the opposite pattern [66]. Somatotropin is lower before calving than post-calving with a peak at parturition [65].

Metabolic Adaptations

In order to transition successfully, the cow must undergo several metabolic adaptations. Between 1 week prepartum and 1 week postpartum, the DMI increase by only 30–50% [67–69] in contrast to abruptly increased requirements for different nutrients. These coordinated processes allow for increased nutrient usage by the mammary gland. The homeorhetic changes required to achieve this tissue mobilization appear to be under genetic control to certain extent only [70]. Data from 113 lactations across 76 cows between 2002 and 2004 revealed that there was no significant interaction between dairy cow strain (Holstein–Friesian) and diet (concentrate supplementation) on milk production, BW and body condition score and lactation profiles. But there was a tendency for a strain and diet interaction suggesting that the effect of concentrate supplementation on BCS gain was, at least partly, strain dependent [71–73].

Dietary energy

During the transition period there is an increased demand of energy for the growth of the foetus and uterus, parturition and onset of lactation [19]. The increased energy requirements can be partly met by increased hepatic gluconeogenesis [74, 75], decreased glucose usage by peripheral tissues, increased FA production from adipose tissues and increased AA mobilization from muscle [30] in cows. In spite of all these changes, the cow goes into NEB. The massive mobilization of body fat is achieved by a combination of increased lipolysis and decreased rates of lipogenesis and FA re-esterification in adipose tissue, leading to the net release of NEFAs and glycerol into the bloodstream [30, 33, 72]. The presence of NEFA in plasma above normal (>0.3 meq/l) indicates that fat is being mobilized in response to NEB. Plasma NEFA begin increasing 3–10 days prepartum, peaking at calving (0.5–1.2 meq/l), and declining postpartum (<0.6 meq/l) in healthy cows [36, 76]. Plasma NEFAs can be (a) efficiently used for synthesis of mammary triglyceride (TG), (b) completely oxidized to carbon dioxide to provide energy for the liver, (c) partially oxidized to produce ketone bodies that are released into the blood and serve as fuels for other tissues or (d) reconverted to storage fat as TG [77, 78]. In addition,

glycerol is a readily assimilated precursor for hepatic gluconeogenesis.

The elevated concentrations of NEFAs often result in significant conversion of NEFA to ketones, e.g. BHBA in liver. The elevated NEFAs concentrations may directly impair neutrophil viability [79] and high BHBA concentrations ($>1200 \mu\text{mol/l}$) in the blood can cause SCK, which is prevalent in around 30% of cows in the first 50 days of lactation [80]. SCK is known to suppress the immune system [81] and acts as a gateway to several other production diseases. Raised blood ketones are associated with a 3–8 times increase in the risk of a displaced abomasums [82]; double the risk of a retained placenta [83]; three times the risk of metritis and six times increase in the risk of developing cystic ovaries [84]. SCK suppressed milk yield by 5.5–7.0% for cows with blood BHBA $\geq 1.4 \text{ mmol/l}$ during the first week after calving [82, 85, 86]. Also fertility is adversely affected, with cows taking on average 22 days longer to return to oestrus [87] and milk production can be reduced by as much as 300 kg in the following lactation [88]. Probably one of the most startling figures is a doubling in the risk of culling in the first 60 days post-calving for affected cows [89]. High rates of these diseases should prompt investigation into the occurrence of SCK in the herd.

Dietary protein

Like energy, there is increased demand of protein during transition period for growth of foetus and uterus, parturition and onset of lactation. A cow can store only 20 kg of mobilizable protein in the body as against 100–200 kg of fat. Thus, the animal can mobilize more fat than protein indicating that it is the protein content that limits production sustained by mobilized reserves. Loss of one kg of weight can provide sufficient energy for about 6–7 litres milk, but will only provide protein enough for 3–7 litres milk. Therefore, a high protein diet should be fed to compensate the gap of three litres milk. Since, the energy during this period comes from body reserves, therefore, there will be shortage of energy in the rumen and the microbes will not be able to utilize any additional RDP. Therefore, for the first few weeks after parturition, the diet must contain high UDP, which should range from 35 to 40% for early lactation and high milk production. Paying close attention to the Aa profile of bypass sources of protein will help supply the essential AAs in the diet. However, balancing UDP alone is not recommended. Adequate RDP is also necessary, so that sufficient ammonia is produced in the rumen to meet the N needs of the microbes. Under normal feeding practices, the complete feed should contain RDP and UDP in 65:35 ratio and 50% of RDP should be soluble protein. The excessive intake of dietary soluble-N can lead to high levels of BUN and in turn high MUN. MUN above 18 mg/dl reflects excessive dietary CP or low rumen degradable non-structural carbohydrates (NSC) such as

starch and soluble sugars, etc. which result in low reproductive performance, higher feed cost, poor milk yield and health problems. MUN below 12 mg/dl reflects low dietary CP and high NSC. Moreover, the CP requirement varies with DMI [90, 91].

Rumen microbiota

Since the diversity, richness and composition of ruminal microbiota are diet specific [92], the fibrolytic or amylolytic bacteria can be manipulated by changing the roughage to concentrate ratio in the ration. A common approach to manage high-producing pregnant dairy cows during the last 3 weeks of the non-lactating period is to shift from a high-forage, low-energy diet to a high-concentrate, high-energy diet [11]. This will increase the population of microorganisms capable of fermenting starch and sugars (NSCs), and hence, produce more SCFAs and propionate. Such an approach also could enhance the development of ruminal epithelium in general and rumen papillae specifically and potentially increase the SCFAs absorptive capacity after parturition when cows will normally take a week or longer to reach appropriate levels of feed intake. A recent study indicated increases in the population of *Lactobacillus* spp., *Streptococcus bovis*, *Megasphaera elsdenii*, *Ruminobacter amylophilus*, *Prevotella ruminicola*, and *Selenomonas ruminantium* [93] after parturition. A decrease in fibrolytic bacteria such as *Ruminococcus flavefaciens*, *Fibrobacter succinogenes* and *Butyrivibrio fibrisolvens* also was detected after parturition. Such responses were primarily driven by the feeding of a high-grain, high-energy diet after parturition, which is a common management practice in the field. Similar results were reported by others, utilizing PCR or sequencing approaches [94, 95] and agree with the characteristics of these predominant bacteria identified *in vitro* [96]. In addition to changes in bacterial populations, Kumar *et al.* [97] and Lima *et al.* [98] reported a decrease in fungi after parturition when a high-grain, high-energy diet was fed and suggested that fungi may play an important role in fibre degradation. Lima *et al.* [98] revealed that *Litostomatea*, a ruminal protozoal taxa, was more abundant with the high-energy diet, but the biological role of this change could not be ascertained.

Nutritional Strategies to Ease out the Transition Phase

Increasing feed intake

It is essential to keep DMI as high as possible during the close-up period and for several weeks post-calving. Aromatic compounds are of particular relevance for cattle since they have almost three times more taste receptor cells than humans do. Many phytogetic or

botanical substances contain sensory-relevant compounds that make feed tastier for cows. These include essential oils, herbs, spices and extracts that provide flavouring properties, particularly aromatic phenolic compounds found in thyme, clove, etc. In a recent trial in the Czech Republic, application of a phytogenic feed additive improved feed and energy intakes (1.2%) in cows. Moreover, milk yield was improved by 5.1% ($P < 0.001$) as was daily milk solids production (fat by 6.7%, protein 1.7%, lactose by 4.8%). The energy corrected milk (ECM) was enhanced by 3.5% [99].

Increasing feed conversion efficiency (FCE)

The second tool to support the freshly calved cow (i.e. immediately after parturition) is to increase FCE by increasing digestibility and by getting more SCFAs and nutrients out of the feed. This can be achieved by supplying helpful rumen microbiota with functional components such as nucleotides and glucans. While a number of yeast products with such components exist, it has been shown that breaking down of whole yeast cells into fragments by autolysis boosts microbial numbers in the rumen. The increase in microbial population leads to improved feed digestibility and increased production of acetate, propionate and butyrate, leading to a better energy balance and higher beef or milk production. A selected phytogenic feed additive and a selected autolysed yeast product support to overcome challenges in feed intake as well as support feed efficiency towards improving animal performance, resulting in more efficient production and greater economic benefit for producers [99].

Increasing density of ration

The NEB or deficiency of glucose during the transition period may lead to various metabolic, physiological, productive and reproductive disorders. To exploit the genetic potential of dairy animals, it is imperative to ease out the transition period by providing the desired nutrients, when these are needed the most. Low-energy and high-fibre diet during the dry period decrease the size of the rumen papillae, which are responsible for the absorption of end products of digestion in the rumen. On the introduction of high-energy diet after calving, the size of rumen papillae increases, resulting in higher and faster absorption of nutrients. But, if a diet rich in soluble carbohydrates and low in fibre is offered four weeks prior to calving, the desired papillae size can be achieved immediately after calving as compared to 6–8 weeks after calving if the diet offered is low-energy and fibrous. The diet of such cows must provide the required 10 MJ ME/kg DMI with 16% CP. However, care should be taken to check excessive intake of high-energy diet in dry period, especially during last 3 weeks of gestation, to avoid complications like difficulties

during parturition, periparturient calf mortality, or affecting adversely subsequent lactation yield.

High producing cows are considered more efficient than low producing cows even though they consume more feed, because more of the consumed energy goes to milk production than maintenance or stated differently: maintenance becomes a smaller proportion of the total energy requirement as energy intake for milk production increases [100].

Singh et al. [101] studied the effect of elevated feeding during –60 days through +120 days postpartum in cross-bred cows fed either as per National Research Council (NRC) feeding standard (Control) or 20% above NRC during 60 days pre-partum (T_1) or fed 20% above NRC during both 60 days pre-partum to 120 days postpartum (T_2) period. During pre-partum period BW gain was improved significantly in T_1 and T_2 groups than that of the control group. The animals fed at higher plane of nutrition took significantly less time for complete relaxation of pelvic muscles; act of calving and for expulsion of placenta than that of control group. Moreover, such cows delivered 2–3 kg heavier calves as compared with normal fed dams. During postpartum period, the average daily milk yield and its composition (milk fat, solids-not-fat and total solids) were significantly higher in T_2 group than that in T_1 and control group. The peak yield was significantly higher in T_2 group, it took longer time to reach peak production, but it was more persistent in this group as compared to T_1 and control groups [101], confirming the earlier reports [102–105], which also observed similar trend in animals fed at higher plane of nutrition than those at lower plane of nutrition. Body weight losses incurred during early lactation were not even compensated by end of 4th month of lactation in C and T_1 groups, whereas the animals in T_2 group gained 2.0 kg. Similar trend in BW loss of dairy cows has been reported by Khatkar et al. [106]. The 1st postpartum oestrus and conception rate were better in high-fed groups (T_1 and T_2) than that of the control group. Yadava et al. [107] also observed earlier exhibition of 1st postpartum oestrus in high-fed animals (48 days) as compared with low-fed animals (51 days). The returns over feed cost of milk production were higher in T_2 group followed by T_1 and control groups indicating the advantage of elevated feeding during pre- and postpartum periods [101].

The net energy of lactation (NE_L) needed in diets of cows and heifers to meet requirements for maintenance and gestation during the transition period revealed that heifers need higher dietary energy due to lower feed intake and the additional energy requirements to support growth [42]. Grain has to be introduced to the cow's ration for at least 3 weeks before the due date and for heifers this should be 5 weeks. The energy density should be between 1.56 and 1.62 Mcal NE_L /kg DM intake [42]. Increasing energy density may stimulate papillae growth and increase acid absorption from the rumen, adapt the microbial population to higher starch diets, increase blood insulin,

decrease FA mobilization from adipose tissue and increase DMI [108].

Increasing dietary NFC or decreasing NDF during the transition period stimulates DMI. When energy density of the diet increased from 1.3 to 1.54 Mcal NE_L/kg DM and CP increased from 13 to 16% at about 3 weeks prior to calving, DMI increased by 30% [109]. Cows fed high NFC (40–42%) diets, consumed more DM during the prepartum period, had lower plasma NEFA, reduced BHBA level and there was a reduction in liver TG [110]. Similar trends were also reported by van de Haar *et al.* [111] and concluded that increasing the energy and protein density up to 1.6 Mcal of NE_L/kg and 16% CP in diets during the last month before parturition improves nutrient balance of cattle prepartum and decreases hepatic TG content at parturition.

Generally, the roughage to concentrate ratio recommended for high yielder is 50:50, but even if best quality silage (11 MJ ME/kg DM) and concentrate (13 MJ ME/kg DM) is offered in above ratio, even then the mixture of these two (12 MJ ME/kg DM) can meet energy requirement up to 55 litres milk/day provided DMI of such feed is as high as 4.3% of BW. Since the DMI is the major constraint, the alternative is either to increase the energy density (by feeding additional maize or adding oil seeds or protected fat) or by reducing roughage to concentrate (R: C) ratio. Formulation of any ration with density beyond 13 MJ ME/kg DM is not possible. It has also been reported that milk production is positively correlated with the intake of DM and water. The commonly used feedstuffs for providing energy to the animal are maize, wheat, sorghum and pearl millet. The particle size of hard coat cereal grains plays a very important role in the efficiency of utilization of nutrients. For the best utilization of nutrients with minimum excretion in faeces the grains shall be crushed and passed through the screen of 1.5–2.0 mm pore size [112]. The total starch contents in the diet of high yielder should be between 20–25%.

Molasses is an easily fermentable source of energy, potassium and sulphur. It is included to increase the palatability as well as level of soluble sugars in the diet. The recommended level of sugars in high yielder is 8–10%. Molasses incorporated up to 6% in the diet have positive effect on the digestibility of nutrients [113]. At higher levels it increases the relative proportion of butyrate and decreases that of propionate in the rumen, which depresses utilization of nutrients and may cause ketosis. The level of sugar and starch beyond 35% in the diet can suppress NDF digestibility [42].

The density of a ration can also be increased by dietary supplementation of oils. But, raw edible oil given beyond certain limits may adversely affect the fibre digestion and bind divalent mineral ions. Even 2% vegetable oil in the diet can affect the microbial population, which results in reduced fibre digestion in the rumen [99]. Therefore the fat should be supplemented in a form that does not affect fibre digestion in the rumen. Bypass fat has a low solubility in the rumen and is not susceptible to bio-hydrogenation in the

rumen. It is digested in the abomasum at acidic pH, without affecting fibre digestion. Different FA oils have been used for making bypass fat (BPF) e.g. palm FAs oil [114, 115], rapeseed acid oil [116] and rice bran FA oil [117, 118] by fusion method.

The concept of 'priming' liver metabolism ahead of calving so that the level of desired poly unsaturated fatty acids (PUFA) versus saturated fatty acids (SFA) are enriched in liver and adipose tissue [69, 119, 120], which reduces the chances of fatty liver by reducing the production of pro-inflammatory cytokines through affecting gene expression in periparturient cows [121–123]. Looor [124] has demonstrated that peroxisome proliferator activated receptors (PPAR) have biological relevance in ruminants and might represent novel biological targets to optimize liver and adipose adaptations to the onset of parturition. Bovine PPAR also responded to pharmaceutical agonists and LCFA [125–126]. Several recent studies have evaluated different types of lipid and their capacity to reduce lipid deposition in liver, but results have been inconsistent.

Manipulating dietary protein

An increase in DMI by 5% can reduce CP requirements by 1%, simply by increasing microbial protein synthesis in the rumen. It has been reported that if the UDP content is increased up to 56% by dietary manipulation, it improves milk yield, its quality (milk protein), and checks loss in weight, first service conception rate as well as pregnancy rate. The key is not only how much protein escapes rumen fermentation, but also the quality of AA content of that protein. There are two ways to increase the UDP content in the diet, one is to include natural protein supplements with high UDP content like corn gluten meal, cotton seed cake, coconut meal and second is to include processed protein supplements (roasted soybean meal or roasted full fat soybeans or formaldehyde treated oil seed cakes). The UDP content of deoiled groundnut cake (DGNC), deoiled mustard cake and deoiled sunflower cake varies between 23–26% of CP, while that of cottonseed cake (CSC), corn gluten meal (CGM) between 46–78% [127]. Lamba *et al.* [128] also observed that the rumen UDP as per cent of total protein in CGM was significantly higher than all other tested protein supplements, it was followed by tomato pomace and was lowest in DGNC. The higher concentration of UDP in the CSC and CGM was mainly because of higher prolamins and glutelins [129]. The degradability pattern of different fractions of proteins indicated that the degradability of globulin was highest, followed by albumin, glutelin and prolamin [130]; the low molecular weight polypeptides in the tested protein supplements showed resistance to rumen microbial fermentation, even after 48 h of rumen incubation [131]. Amongst the cereal and millet grains oats have the lowest UDP value followed by wheat < barley < waste bread < maize and sorghum [132].

Efficient utilization of protein supplements would not only improve the productivity and reproduction of the animal, but may also help in reducing the global warming by decreasing the enteric methane production especially from animals fed low quality roughages. Bakshi et al. [133] showed that the feedstuffs which have high UDP content had low methane production potential. Use of such feed ingredients should be advocated to formulate the ration for high yielders to sustain the livestock productivity. Each ingredient should be included with a specific purpose, and diet should be formulated to meet the specific targets. The efficiency of rumen fermentation depends upon the amount of microbial biomass synthesized, which in turn depends on the synchronization of protein and carbohydrate breakdown. The UDP level has a profound effect on the availability of N for rumen microbes. Low UDP will not only hamper the performance of the animal but also lead to loss of ammonia from the rumen. *In-vitro* studies revealed that diet containing ME at 110 or 120% of NRC with UDP (24% of dietary CP) supplemented with niacin gave the best response as far as digestibility of nutrients and availability of ME was concerned [134].

Increasing dietary minerals

Cow's milk is a good source of calcium, chloride, potassium, magnesium and sodium. Most of the B-complex vitamins are present in milk. A marginal deficiency of the vitamins and minerals may not have significant impact on milk quantity and quality. However, calcium is one of the crucial elements in the ration to be considered more carefully in transition cows. At the beginning of lactation, the sudden demand of calcium for milk production increases dramatically, leading to fall in blood calcium levels and the condition is called hypocalcaemia. This stimulates the secretion of parathyroid hormone (PTH), which stimulates the bone resorption [51]. It takes 2–3 days for the PTH cycle to become fully functional [135]. It also activates vitamin D₃, which increases absorption of Ca from intestine and mobilized bone Ca. But this process requires 24–48 h, and cannot prevent animals from milk fever/parturient paresis as more than 60% of cases of milk fever occur within 24 h of parturition. To avoid incidences of milk fever, the best feeding management practice is to provide low Ca (<50 g/day) diet during last 2–3 weeks of gestation, which should be increased to ~100 g/d at least two days before parturition. The diet, after parturition, should have sufficient Mg, an essential activator of vitamin D₃ in liver. The anionic diet (–12 meq/100 g DM) resulted in zero incidence of mastitis and milk fever as compared with 14.3% in cationic diet (15 meq/100 g DM) fed cows. The incidence of metritis was also zero against 71.4% in cows fed cationic diets. In animals fed diet containing anionic mineral mixture (calcium chloride 33.4, magnesium chloride 33.3, sodium chloride 18.3, magnesium sulphate 8.3 and calcium

hydrogen phosphate 6.7%), the 4% fat corrected milk (FCM) production was increased by 1.13 kg during the 90 days postpartum period. About 90 g anionic mixture is recommended per day for 3 weeks pre-partum [136].

Metabolic acidosis caused by a negative dietary cation–anion difference (DCAD) favours mobilization of calcium from bone, whereas high dietary potassium concentrations and positive DCAD suppress this process [137]. The subclinical hypocalcaemia is believed to be a contributing factor in disorders such as displaced abomasum and ketosis [51]. Hypocalcaemia also leads to increased secretion of cortisol, which is considered to be a factor in increased incidence of retained placenta [138].

Manipulating dietary fiber level and particle size

For optimal animal production, the rumen should operate to its maximal efficiency in terms of mixing, rumination and emptying. Continuous mixing of rumen contents improves the intimacy between ingested feed particles and the microbial population, essential for optimal fibre digestion. Active bouts of rumination support feed breakdown and promote saliva secretion, which assists the control of rumen pH. But for effective rumen mixing and rumination, care must be taken to ensure sufficient fibre level with optimum physically effective fibre (PEF) length in the total mixed ration (TMR). Complete feed of high-yielding cow should not contain less than 21% ADF or 28% NDF [42]. The forage DMI as per cent of BW should be 1.4–2.4% [139]. Maximum individual intake of NDF by high-yielding cows reaches up to 1.3–1.4% of BW. The forage NDF intake as a per cent of bodyweight should be about 0.8–0.9 [140] or it should not constitute more than 17–18% of total DMI. But it can be increased to 30–34% by using wheat bran, orange peel or citrus pulp, rich in soluble fibre, i.e. pectins. Citrus pectins are easily and extensively degraded, producing acetic acid, which is less likely than lactic acid to cause a pH drop and result in acidosis [141]. Due to high fibre content in citrus pulp, the long rumination of citrus pulp produces large quantities of saliva that has a buffering effect on rumen pH. Citrus pulp is therefore considered as a safer feed than cereals for animals fed high-concentrate, low-roughage diets in high-yielding dairy cows [142]. The concept of high milk yielders in developed and developing countries differs with respect to milk yield. For instance, in the former case, animals yielding 30–40 litres/day are considered as high yielders, but in the latter case those yielding 25–30 litres/day are considered as high yielders. Accordingly the recommended R:C ratio varies. In developing countries, the R:C ratio, should be 50:50, 60:40 and 70:30 in early, mid and late lactation; and in developed countries the R:C ratio during early lactation should be 40:60. In developed countries, the roughage portion is generally made up of alfalfa forage supplemented with corn silage, and if the green forage is non-leguminous then it is supplemented with grass hay, but fillers such as

straws and stovers are not incorporated in the diet of high yielders. In developing countries such as India, leguminous forage such as berseem (*Trifolium allexandrium*) is mixed with small quantity of wheat/rice straw and fed to the animals. Straws/stovers with improved quality after processing with urea could replace hay up to 25% (DM basis), without any adverse effect on productive performance of such animals. Naturally fermented wheat straw with urea (96.5:3.5) moistened to 40% and stacked for 9 day (FWS) could meet the basal energy and protein requirement for maintenance of adult ruminants [143, 144]. A 396-day lactation trial revealed that *ad lib* feeding of FWS with low-protein concentrate resulted in higher milk production without any adverse effect on the quality of milk and conception rate in buffaloes [145]. The improved milk production could be because of the low enteric methane production from FWS [146].

Rumen pH is stabilized by a proper balance of slowly (fibrous) and rapidly fermentable NFC and optimum PEF length to stimulate rumen mixing, rumination and saliva production [147]. Low rumen pH caused by shorter length of PEF reduces production efficiency by increasing rumen emptying time, reducing rumination, decreasing DMI, impairing rumen contractions, fibre digestibility, and microbial yield from the rumen and milk fat depression [147, 148]. If the degree and/or duration of low ruminal pH are substantial, it leads to health problems secondary to subacute ruminal acidosis such as rumenitis, laminitis, displaced abomasum and liver abscesses [149]. Coarse, fibrous feeds stimulate more saliva production during eating and stimulate rumination than do finely ground feeds or fresh pasture [150]. Beauchemin *et al.* [151] fed small amounts of cereal straw (<0.5 kg/day) as a source of structural fibre and of suitable length (4–8 cm) and structure was fully incorporated into mixed rations, encouraging animal performance results were noted. The size of these particles is eventually reduced by rumination. When particles become small enough, i.e. less than about 12 mm [152], they pass out of the rumen through the reticulo-omasal orifice. Coarse fibre particles in the rumen may also act as a sieve to prevent larger particles from passing out of the rumen [152].

It has been reported that reducing forage particle length (<6 mm) may reduce feed bunk sorting behaviour of dairy cattle. Rations containing a greater proportion of longer particles (>19 mm) are likely to have a larger difference between the feed originally offered and that consumed throughout the day because of sorting activity [139, 140, 153, 154]. Forage/silage/hay particles length (shorter 6 mm or longer 19 mm) can affect the production and health of dairy cows. The long particles can result in poor silage fermentation [155]. The animals fed such forages/silage/hay spend more time in rumination and are retained for more time in the rumen [156]. The shorter forage/silage particles can increase the risk of subacute ruminal acidosis [157, 158]. Considerable research has been conducted to explore the impact of forage/silage/hay particle length on

rumen function and milk production, but the results have been inconclusive. Some studies have shown that reducing the particle length of alfalfa hay/silage, barley silage and corn silage does not affect rumen pH [159, 160], whereas others have demonstrated a reduction in rumen pH [161–163]. Studies have also shown that reducing the forage/oat silage particle length can increase DMI [153, 160, 164, 165], others reported that reducing the particle length of alfalfa silage had no effect on DMI [159, 162, 165] or reduce DMI [166]. Reducing the particle lengths of alfalfa hay/silage and corn silage can lower milk fat percentage [162, 166, 167]. Other studies did not observe any effect on milk fat percentage [153, 162, 164, 165, 168]. Similarly, long PEF increased milk yield [169], while others did not observe any effect on milk yield [153, 163, 165, 170]. The disparities among studies may be caused by interactions between numerous factors, including reduced forage particle size, grain source, forage source, forage to concentration ratio, range of particle lengths and forage NDF content. Therefore, proper ruminal fluid pH appears to be the result of interaction between measured chemical fibre adequacy, i.e. (level of NDF), ADF in the ration and particle length of PEF. A survey of 1086 dairy farms in France and UK reported an additional 1.84 litre milk/cow/day within 12 months of adopting integrated feeding system to optimize chemical and physical characteristics of ration, while feed consumption fell by 0.74 kg DM/cow/day, with a 12% improvement in FCE (1.276 versus 1.141 litres of ECM/kg feed DM). By improving FCE, methane output per litre ECM or per kg weight gain is reduced by 20% of current levels. Adoption of the system for dry cows showed improved animal health, with the incidence of health issues around calving reduced from 45 to 16 cases per 100 cows calved [171].

Feeding management

The diet should be offered as TMR in order to avoid the selective intake of feedstuffs, effective utilization of nutrients and to save labour thereby economizing production. The feed intake is correlated with ambient temperature. The heat associated with digestion of feed, peaks about 3–4 h after feeding. By feeding cows in early morning (5–6 a.m.), the heat of digestion peaks at 8–9 a.m., and allows the cow to dissipate some of that heat before the day gets hot. A cow fed at 8 a.m. will have her peak of heat production at 11 a.m. to noon when the day is hotter, which is undesirable. Similarly, cows fed late in the evening will be more comfortable and likely to consume feed and their peak of heat production will occur during the night, when environmental temperature is low. In addition, frequent small meals (3–4 times a day) result in less heat generation than fewer, but larger meals [113].

Water is the most important nutrient for the cow. DMI and milk yield depend on free access to fresh, clean and cool water. Water should be provided in a well-lit area

within 15 m of the feed manger. About 5 litres water is required per kg of milk produced. Decrease in water intake by 40% results in 16–24% decline in DMI and considerable decline in milk yield [172].

Feed additives supplementation

In order to maximize profitability from the dairy farming, farmers are increasingly focused on improving the efficiency of feed conversion. One method is the use of feed additives to optimize health and well-being. Many feed additives have now become routine components of transition cow rations, in spite of the fact that each one addresses only one particular aspect of the metabolic and/or immune challenges of the periparturient cow. Therefore, combinations of additives are offered and often included in the transition diet. These aim at combining beneficial effects on energy metabolism with stimulatory action on the liver and immune system [173]. Wadhwa *et al.* [2] indicated that strategic supplementation can have a large impact on the increase in animal productivity and decrease in environmental pollutants in the non-intensive systems; and this large change is induced through optimization of the rumen function.

Inflammation modulators

Approximately 50% of cows in dairy farms tend to produce less milk, lose more body reserves, and become less fertile because they suffer from inflammation, which may not be clinically evident [56]. But can be diagnosed via acute phase protein changes, i.e. increase in haptoglobin and reduction in albumin, lipoproteins and retinol binding protein (less liver synthesis of usual proteins). It seems unrealistic to expect an elimination of any inflammatory process. Therefore, it also seems advisable to modulate the inflammation phenomenon *per se* and to attenuate the systemic response to it. There are several possibilities for modulating inflammation by using anti-inflammatory drugs like acetylsalicylic acid, which had positive effects on milk yield and fertility [174]. Some PUFAs, such as omega-3(ω -3) and conjugated linoleic acid (CLA), have also been found to be effective. Lately, attention is being paid to herbal feed additives, which have anti-inflammatory or immune-stimulatory properties, and consequently might be used as nutraceuticals.

Omega-3 fatty acids (ω -3)

These FAs have been well known as anti-inflammatory nutrients in humans since 1980 and Bertoni *et al.* [175] showed that their levels in blood markedly decrease around calving. Rumen-protected and rich in eicosapentaenoic and docosahexaenoic acids (EPA and DHA) were given either alone (16.4 g/cow/day of ω -3 from fish oil; [175]) or in combination with vitamin E (18.5 g/cow/day of ω -3 from algae-extract; [176]), during the peripartum increased plasma concentrations of ω -3. All cows involved in the

experiments (fed ω -3 or non-treated) suffered an enhanced inflammatory condition immediately after calving, highlighted by the rise of positive acute phase proteins (haptoglobin). However, the cows fed ω -3 were able to recover better, as confirmed by a quicker recovery of the negative (albumin, lipoproteins) and by the fast reduction of the positive (haptoglobin) acute phase proteins after calving. Interestingly, ω -3 administration also exerted a positive effect on the energy metabolism.

Conjugated linoleic acid

This FA is known to reduce milk fat content in dairy cows and to attenuate inflammatory phenomena in animals. To confirm the latter effect on dairy cows during their transition period, 20 g of a mixture of CLA isomers (50% of c9t11 and 50% of t10c12 protected from rumen) was given orally from -28 through +28 days postpartum. The CLA feeding resulted in an attenuation of the inflammatory consequences after calving, as suggested by the quicker increase of some negative acute phase proteins (improved liver synthesis). Moreover, CLA treatment was able to reduce the NEB, as confirmed by several indices showing an increased energy availability (increased feed intake and plasma glucose) and a reduced energy expenditure (slightly lower milk yield and fat content, lower NEFA and BHBA in plasma) [56].

Antioxidants

Oxidative stress (OS) arises due to an imbalance between pro-oxidants and antioxidants, which can occur under circumstances of increased antioxidant utilization, impaired buffering capacity or immune function. OS seems to play a central role in the regulation of the metabolic activity of some organs and productivity in farm animals [177, 178]. For example, during the last trimester of pregnancy and at the beginning of lactation the rapid foetal growth and the production of large amounts of colostrum and milk, an increase in both maternal and foetal metabolism result in increased requirements for micronutrients, including antioxidants [179]. Therefore manipulation of micronutrients and antioxidants has the potential to control the effects of oxidative damage, such as occurs during the transition period [180]. The relationship between OS and disease and metabolic disorders is further demonstrated by lowered antioxidant status during mastitis, metritis, retained placenta, acidosis, ketosis and milk fever [177].

Selenium and vitamin E represent ideal antioxidant supplements to improve ruminant health, production and reproduction. Finally, as OS has detrimental impacts on immune function and associated health disorders, future research need to focus on the maintenance of redox homeostasis and interactions between antioxidants for metabolic use especially during the transition period. Furthermore, the use of plants and plant parts such as yerba mate (*Ilex paraguariensis* A. family: Aquifoliaceae) and moringa leaves that are rich in antioxidants is an emerging area of work in livestock nutrition and has potential to

enhance animal welfare, production and health [181–183]. Exploitation of these novel resources in animal diets needs to be explored.

There is strong evidence that supplementation of the diet with vitamin E and/or α -tocopherol, selenium, vitamin A or retinol and/or β -carotene and vitamin C is a practical way of enhancing immune-cell function and decreasing inflammation [184]. The supplementation of rumen-protected ω -3 PUFA plus vitamin E from –21 through +21 days relative to parturition lowered postpartum blood concentrations of NEFA, BHBA, bilirubin and increased concentration of α -tocopherol, indicating an overall improvement in liver function and attenuated inflammatory response postpartum [176]. Fresh forages, such as perennial ryegrass, may therefore be beneficial in the peripartum period, being high in ω -3 rich oils, vitamin E and having a fast rumen-passage rate, ensuring minimal biohydrogenation of the desirable fat. These advantages must be weighed against the high dietary RDP and dietary potassium, both of which may have negative consequences for the health of dairy cows.

Clinical and sub-clinical mastitis remain among the biggest drains on dairy profitability. Since vitamin E cannot be synthesized by the cow; supplementing optimal levels of vitamin E in pre-calving and immediate post-calving rations has proven to increase immune system resistance to mastitis infection, reducing mammary infections at calving by up to 80%. Supplementing rations with vitamin E has proved cost-effective and efficient in combating this serious problem. It decreases the duration of mastitis in infected cows by as much as 50%, as well as cases of udder oedema and milk somatic cell counts [185].

β -carotene

Supplementing β -carotene in the ration is advisable because it is not always available in optimum amounts in dairy cow feed, e.g. maize silage has a negligible concentration of β -carotene. In addition, its presence in most grass silage is below the requirement for optimal cow fertility. A longer productive life is associated with low culling rates and high fertility. β -carotene, a plant pigment acts as a vitamin to support fertility in dairy cows. It accumulates in the ovaries and participates in the synthesis of the hormones oestrogen and progesterone, both of which are important for ovulation and for safeguarding the embryo. Trials conducted worldwide show that optimal dietary levels of β -carotene increase conception rates and decrease the number of abortions. Pregnancy rates for first inseminations in a practical farm trial increased by almost 40% in β -carotene supplemented cows. Calving intervals were reduced by 27 days and milk yield increased by more than 400 kg per cow and lactation. Low dietary levels of β -carotene often mean that the oestrus is hard to identify, which can result in missed breeding opportunities. One missed oestrus means that 21 days are lost before the next lactation starts and several missed oestrous can result in otherwise healthy cows being culled. Statistics

collected by the USDA [186] reveal that around 26% of cows culled from modern dairy herds are slaughtered because of fertility problems. Correct supplementation of β -carotene in rations leads to an improvement in cows' fertility, health and welfare. Other recorded advantages include more regular ovulation, better foetal health and reduced incidence of retained placenta; as well as better calf health [187].

Bypass fat

In developing countries, the ration of crossbred cows and high-yielding buffaloes during late gestation and early lactation is often energy deficient and do not meet the requirements, thus forcing mobilization of body fat reserves and the animal goes in NEB [188]. Under field conditions crossbred cows and buffaloes often lose 80–100 kg BW after calving [115]. Such debilitated animals do not come in heat, unless the lost BW is fully or partly recovered. But at a university dairy farm the loss was considerably low, e.g. lactating crossbred cows fed as per NRC feeding standard exhibited loss in BW (–9.28 kg), while those fed at 120% of NRC feeding standard gained in BW (+4.2 kg) after 8th fortnight postpartum. The loss/gain in BW after calving depends on the level of feeding [101]. Fat plays important role in the performance of lactating animals. BIS [189] have recommended only 2.5 and 4.0% fat in the compounded concentrate mixture, whereas NRC [42] has recommended 3.0% fat in the complete feed for dairy cattle. However, reports indicated that dietary fat up to 5% of DMI have direct positive impact on the quantity as well as quality of milk produced, but beyond this level depression in milk yield has been recorded. The BPF could be incorporated up to 13% in the diet with continual increase in milk production [190].

Impact on milk yield and composition

A number of studies have revealed that dietary supplementation of BPF in crossbred cows and Murrah buffaloes during the transition phase/early lactation improved the milk yield and fat content significantly [191–194]. Further, supplementation of BPF in the ration of cows and buffaloes increased the proportion of unsaturated and long-chain FAs of milk fat [195–197]. An on farm trial on 24 pregnant buffaloes in their 2–4 lactation from –30 through +120 days postpartum revealed that the 6% FCM yield, solids corrected milk (SCM), ECM and fat per cent were significantly ($P < 0.01$) higher in BPF supplemented group as compared with control group [198], confirming the earlier reports [199–205]. Field trials conducted by Wadhwa *et al.* [118] revealed that the average daily milk yield of crossbred cows in early lactation was improved in the BPF supplemented group as compared with the control group. Also BPF supplementation improved the composition of milk, i.e. protein, lactose and solid not fat (SNF) content in milk. The BW of the animals was improved considerably in BPF group as compared with control group. Another trial on

high-yielding crossbred cows in early lactation revealed that supplementing the bypass fat with either bypass protein or chelated Cu and Zn did not have any beneficial effect on milk yield or its composition [206].

Effect on reproduction

A feeding trial was conducted to assess the impact of supplementing BPF at 2.5% of DMI on the performance of multiparous crossbred cows from -40 days through +90 days extended up to 210 days postpartum. There was increase in birth weights of the calves, while time taken for expulsion of foetal membranes, involution of uterus, onset of cyclicity, the service period and number of inseminations per conception were reduced ($P < 0.05$) in BPF supplemented group [193]. The concentration of circulating cholesterol, the precursor of progesterone secreted by the corpus luteum, which helps in preparing the uterus for embryo implantation and further maintenance of pregnancy, was also higher in the BPF supplemented group. These findings were confirmed by different workers in crossbred cows and buffaloes [198, 204, 207–210].

Protected AAs

The supply of both RDP and UDP should meet the requirements of essential AAs for optimal animal performance. It is confirmed that Met and Lys are the first limiting AAs in milk production [42]. The ideal ratio between Lys and Met should be 3:1 to maximize milk protein output. One of the newest technologies that can help achieve this balancing strategy is rumen-protected AAs. The benefits of AA balancing include improved dairy performance, improved fertility, reduced greenhouse gas and overall profitability [211]. Preynat et al. [212, 213] reported that feeding rumen protected Met (RPM) did not affect milk yield, but increased milk CP (2.94 versus 3.04%). Supplementation of Met, histidine and Lys can have substantial effects on milk protein content and yield [214]. Osorio et al. [215, 216] support the hypothesis that feeding RPM enriched diet during the transition period is beneficial for cows, confirming the earlier report [217]. Socha et al. [218] fed cows either 10.5 g RPM/day or 10.2 g RPM/day and 16.0 g/day of rumen protected Lys (RPL) 14 days prior to the expected calving through early lactation. Cows fed RPM plus RPL produced more milk during early lactation than those fed RPM alone.

Protected Met, Lys and choline

It has been reported that as much as 30% of the Met absorbed by ruminants is used for choline synthesis [219]. Sai et al. [220] studied the combined effect of dietary supplementation of RPM, RPL and rumen protected choline (RPC) on the performance of crossbred calves. Supplementing RPM, RPL and RPC increased the FCE, plasma TG, very low-density lipo-proteins (VLDL) and average daily gain; and reduced BUN and NEFA levels significantly in crossbred calves. Amrutkar et al. [221]

also assessed the combined effect of supplementing ration with RPM, RPL and RPC on the performance of pre-parturient crossbred cows in 2nd to 4th lactation. BCS on the day of parturition, duodenal supply of Met and Lys, plasma TG, VLDL and phosphatidyl choline levels were higher in RPM, RPL and RPC supplemented groups, while the vitamin E level was higher in RPC supplemented group, on the day of parturition, indicating better immune status, confirming the earlier reports [222, 223].

L-carnitine

Carnitine is an AA derivative and nutrient involved in lipid (fat) metabolism in mammals and other eukaryotes. Its exogenous source is through the diet and endogenously it is synthesized by Lys and Met. L-carnitine has two essential functions, firstly it is the only physiological carrier for transport of FAs inside the mitochondria where they are burnt to produce energy (β -oxidation) and secondly it removes acetyl groups from the mitochondria, which are formed by lipid and amino-acid metabolism. L-carnitine prevents accumulation of FAs in tissues (anti-steatogenic action) and removes acetyl groups, avoiding the formation of ketone bodies (anti-ketogenic action). It improves muscular function and favours the contractile activity of cardiac muscle and increases coronary flow. Deficiency of L-carnitine may lead to ketosis, hepatic steatosis, muscular asthenia and atony and cardiac insufficiency.

Vitamins

Niacin

It plays an important role in the metabolism due to its incorporation into the coenzymes, NAD and NADP [224], which are required in most of the biochemical reactions in the animal system. Both forms of niacin, i.e. nicotinic acid (NA) and nicotinamide (NAM) can be converted into the coenzymes, although they contain only NAM as a reactive component. Apart from feed as a source of niacin, nearly all species of rumen microbes synthesize this vitamin [224–227] from tryptophan [228] and quinolinate [229], but rumen ciliate protozoa cannot synthesize and they obtain niacin from bacteria or feed [230]. The synthesis might vary with feeding regimens. Niacin synthesized in the rumen is adequate for optimum performance of low yielders [231], but may be inadequate for high milk yielders [232]. However, a general supplementation cannot be advised [42, 233]. The increase in microbial protein production after niacin feeding was considered responsible for enhanced milk production [234–237]. However, [238] observed a significant decrease in milk protein concentration, after niacin supplementation.

A meta-analysis of 27 feeding studies involving NA supplementation to dairy rations showed no improvement in lactation performance when NA was given at 6–12 g/day [42, 239]. Supplementation of NA at high doses to the periparturient dairy cows has given inconsistent results.

Jersey cows supplemented with 48 g/day of free NA from 30-day prepartum until calving had lower levels of plasma NEFA at calving and less DMI decline during the last week of gestation [240]. However, these results could not be reproduced when both Holstein and Jersey cows were supplemented with up to 98 mg NA/kg BW/day, from 30-day prepartum to 21-day postpartum [241]. Supplemental niacin is extensively degraded or transformed in the rumen. Absorption of NA through the rumen is probably insignificant [242–244] and only 6.7–17% of supplemental NA was estimated to reach the duodenum in the bovine [244]. Kumar and Das [245] did not observe any significant beneficial effect on the growth and nutrient utilization in buffalo calves fed diet supplemented with 100 or 200 ppm niacin.

Choline

Supplementation of choline, a component of phospholipid and methyl donor facilitates lipid absorption and transport, thereby favouring milk fat synthesis. This also helps in improving milk production. Choline present in feedstuffs is degraded rapidly by the rumen microbes [246]; and cows are unable to synthesize sufficient choline to match the level of body fat mobilized during early lactation. This results in accumulation of fats in the liver, resulting in fatty liver syndrome and affecting 50–60% of cows at parturition or shortly after calving. The accumulation of mobilized fat in the liver also means the cow's NEB is not corrected, so more body fat is mobilized and the liver becomes even further overloaded in a cascading cycle. The key is to boost choline supply and support liver function by feeding RPC supplement (20–60 g/day) for 3 weeks either side of calving, which resulted in increased daily milk yield, and a reduction in post-calving health disorders, such as retained placenta, metritis, ketosis and mastitis. But RPC supplementation did not affect fat or protein concentrations in milk, or plasma levels of glucose, BHBA, cholesterol, NEFA and liver TG significantly [247–252]. However, around parturition, NEFA concentrations in plasma were lower in treated animals than in controls, suggesting improved lipid metabolism as a result of RPC supplementation. Zom *et al.* [253] reported that supplementation with RPC did not affect blood metabolites but decreased the liver TG content during early lactation; further examination [254] of the changes in gene expression in the liver from this study suggested that RPC supplementation resulted in increased expression of genes related to processing of FAs and VLDL assembly. RPC supplementation also increased α -tocopherol plasma concentrations, suggesting improved immunity in dairy cows. Gupta [255] revealed that RPC supplementation (30 g/animal/day) during –21 through +10 days extended to 90 days postpartum improved the postpartum DM intake, nutrient digestibility, blood glucose and milk yield, while plasma TG, loss in BW and BCS were reduced irrespective of the pre-calving dietary energy level. Garg *et al.* [114] supplemented the diet of lactating crossbred

cows with either 100 g BPF/animal/day or 100 g BPF and 10 g RPC/animal/day for 90 days. The daily milk yield and fat content, PUFAs, total unsaturated fatty acids also increased significantly ($P < 0.01$) as compared with control group. NEFA in the blood serum were reduced by 16.12 and 24.19% ($P < 0.01$) in groups II and III, respectively, confirming the earlier reports [249, 256, 257]. Blood glucose and urea N were not affected by the dietary treatments.

Biotin

Some biotin is synthesized by rumen microbes and high yielders respond well to extra biotin in their rations. This is due to the fact that the natural biotin synthesis by the cow decreases when high-energy diets are fed. A summary of five published studies in the USA and Europe indicated that daily supplementation of 20 mg biotin/cow/day boosts milk yield by more than 2 kg/day in high-yielding cows (36 kg/cow/day). Biotin has a proven ability to stimulate glucose synthesis in the liver, which is the main energy source for milk production. Biotin is also a crucial component in the synthesis of keratin (strong horns and robust hooves) and lipids. Results from independent trials worldwide prove that just 20 mg biotin/day significantly ($P < 0.05$) reduces incidences of the most common hoof disorders in dairy cows, such as sole ulcers, white line disease, sandcracks and digital dermatitis [187]. Preventing lameness in this way not only produces an economic advantage for biotin supplementation, but it also plays an important role in enhancing animal welfare in the dairy herd.

The feeding of high concentrate diet during early lactation reduced rumen pH, which in turn reduced the biotin synthesis [258], leading to laminitis, which adversely affected milk production. The biotin supplementation reduced claw lesions and also improved milk production and reproductive performance [259]. *In-vitro* gas production studies revealed that supplementing the complete feeds with niacin, biotin or vitamin E individually did not have any significant effect, but niacin and vitamin E combination significantly improved the net gas production, digestibility of organic matter and ME availability as compared with control. However, addition of biotin did not show any additional benefit in improving the digestibility of nutrients and availability of ME [260]. *In-vivo* trial on 24 crossbred cows from –40 through +110 days postpartum revealed that the dietary combinations (ME 100, 110 or 120% of NRC and UDP 24% of CP or ME 120% of NRC and UDP 40% of CP supplemented with biotin, niacin and vitamin E) did not have any significant impact on the daily milk yield. It was concluded that the diet containing ME at 120% of NRC with 40% UDP of dietary CP supplemented with niacin, biotin and vitamin E decreased ($P > 0.05$) BUN, increased ($P > 0.05$) milk yield and increased ($P < 0.05$) lactose content in milk in high-yielding crossbred cows, but UDP beyond 24% of CP did not have any significant effect on milk yield [261].

Glucogenic precursors

Propylene glycol (PG)/glycerol

The oral drenching [262] or top dressing [263] of TMR with PG has been used in prevention and treatment of ketosis and for increasing milk yield in cows [264, 265]. PG is rich in energy (4.7 Mcal NE_L), and is easily and rapidly absorbed and metabolized in the rumen [266]. Roughly 50% can be metabolized within 1–2 h after feeding, and 80–90% usually metabolized 3 h after feeding. PG improved available energy during the transitional period of cows, offering beneficial effects, especially in environments of high temperature and humidity. PG is converted to propionic acid in the rumen [267–269] and transported to the liver, where it is converted to glucose through the gluconeogenesis [270–273]. PG addition in the diet lowered NEFA and BHBA, increased insulin and glucose levels in blood [264, 274–279]. Incorporation of glycerin and PG in TMR resulted in higher blood glucose levels, milk yield and BCS [280–283].

One of the main goals when feeding dairy cows is to maximize microbial protein production, because microbial protein is of high quality and is an important source of MP that is used for milk production [42]. But, Clapperton and Czerkawski [271] and Ivany and Anderson [284] indicated that PG might have inhibitory effects on rumen microbes. Chibisa et al. [285] revealed that PG supplementation as a top dress on basal ration from day –7 through +45 days postpartum had no effect on total urinary purine derivatives excretion; consequently, microbial N supply and microbial efficiency was not affected. Results indicated that supplementing PG over the transition period had no effect on microbial protein production in the rumen.

Propionate

Its supplementation as Ca propionate or other mineral salts has been suggested as a strategy to enhance supply of glucogenic precursors. However, results have been variable and often disappointing, possibly due to the relatively low levels of supplementation-especially relative to normal rumen propionate production [286]. van Kneusel et al. [17, 287] suggested that ration formulations which reduce the lipogenic-to-glucogenic nutrient ratio would reduce acetate and butyrate and increase propionate concentration, improve the negative energy balance (NEB) and decrease plasma ketone concentration in early lactation.

Combined effect of glucogenic additives

Feeding a dry energy supplement (45% beet pulp, 22% sugarcane molasses, 17% PG and 16% calcium propionate) 21 days either side of parturition, improved DMI, milk yield, milk fat and protein contents, especially at low level, without any effect on incidence of ketosis. The glucogenic additive (450 g/head/day) comprising calcium propionate and loose PG (1:1) fed 2 weeks before calving and till 6 weeks postpartum did not affect DMI; however, it

improved DM and CP digestibilities and increased daily milk yield by nearly 5% [288].

The NEB can result in poor reproductive performance, such as prolonged postpartum anestrus, lower progesterone production by the corpus luteum, and low rates of conception [32, 289]. NEB and loss of BCS in the early postpartum period affected follicular development [290] and reproductive performance [289, 291–293]. Kim and Suh [27] also found that the number of days to first breeding after calving was longer in cows with marked BCS loss than in those with moderate BCS loss. In addition, Miyoshi et al. [266] found that transition cows fed PG have an earlier oestrus onset, because insulin is important in normal ovarian function. Lien et al. [265] reported days to first oestrus (77.44 versus 63.88 days), and placental retention (25 versus 8 heads) were reduced, and the pregnancy ratio was higher in the PG group than in the control group. Cows fed glycerol and Ca-propionate, which are glycogenic agents, had a shorter interval [92 versus 103 days, ($P < 0.05$)] between calving and last insemination [294]. It showed beneficial effect on cow reproductive indicators, i.e. reducing the time interval from calving to successful insemination and improving insemination index [295].

Dietary buffers

The diet of a high-yielding cow contains about 50% concentrate mixture containing 40–50% cereal grains result in less salivation and thereby reducing rumen buffering capacity. Feeding such concentrates leads to increased acid production resulting in decline in rumen pH. As rumen pH drops to near 6, fibre digestion is sharply reduced, the animal goes off feed, and DMI is reduced. Feeding concentrates depresses not only the efficiency of feed utilization, but also decreases milk fat percentage. The dietary buffers such as sodium bicarbonate, magnesium oxide and calcium carbonate etc. can be used to neutralize the acidity in the rumen. Addition of dietary buffers at the rate of 1.5% in the high concentrate (cereal based) ration can increase DMI, milk yield and its fat content. Feeding a properly balanced ration containing high-quality forage like alfalfa, which has a high buffering capacity, can minimize the need for buffers [296].

Improving gut health with eubiotics

Eubiotics are a new range of feed additives used to improve gut health in monogastrics, which have also been shown to modulate the rumen microflora. Precisely these are blend of essential oil components, which are steam volatile, plant secondary metabolites. In this sense, the term 'essential oils' is misleading, because they are neither indispensable nor are they fats or oils; but they are polyphenols or terpenes. Crina Ruminants, a precise blend of essential oil

components, increased milk yield on average by 1.5 kg/day when top-dressed the TMR. The modulation of rumen bacteria by essential oils can lead to more than 7% increase in milk production [187]. The essential oils stimulate animal appetite, feed consumption and optimize the rumen environment to ease the transition phase. This, in turn, allows for the optimum production of milk and its protein and fat components.

Direct-fed microbials/probiotics

According to Food and Agriculture Organization of the United Nations (FAO) a probiotic can be defined as a 'live microorganism which, when administered in adequate amounts, confers a health benefit on the host'. The beneficial effects of probiotics are manifested by regulation of intestinal microbial homeostasis, stabilizing the gastrointestinal barrier function, secretion of bacteriocin, immune-modulatory effect and anticarcinogenic enzymes, interference in pathogen colonization, etc. [297, 298]. The probiotic organism can be of prokaryote (bacteria) or eukaryote (yeasts) origin. Amongst the nine genera of prokaryote (bacterial) probiotics, *Lactobacillus* is one of the most tested and successful probiotic class that gained greater application in both human and livestock. As for *Lactobacillus*, the genus *Enterococcus* has been noticed as natural inhabitant in the gastrointestinal tract of animals [299]. While among the three genera of eukaryotic (yeast) category of probiotics, *Saccharomyces cerevisiae* and *Aspergillus oryzae* are the most commonly used ones in ruminants [300]. These are also called 'direct-fed microbials' (DFM). The DFM enhance the microbial fermentation and thus increase substrate utilization, e.g. stimulate the fibre digesting and lactate utilizing bacteria and reduce the transient postprandial drop in pH. In addition, antioxidant extracts stimulate the growth of ruminal fungi that have been shown to play an important role in fibre digestion [301]. The DFM improve overall energy supply and metabolic profiles especially if propionic acid production is enhanced [302]. An increase in relative proportion of propionic acid will stimulate gluconeogenesis. Improvements in levels in blood glucose, and decrease in NEFA and BHBA have been observed [303] confirming the potential positive effect on energy balance, especially postpartum. Feeding of diets supplemented with DFM increased DMI, total VFA production and milk production in transition cows [303, 304].

Dietary supplementation with 5×10^9 cfu of *Enterococcus faecium* and 2×10^9 cfu of *S. cerevisiae*/cow/day from -21 through +70 days postpartum, increased milk yield in cows [303] and in lactating Saanen dairy goats [305]. Dietary supplementation with a combination of *L. acidophilus* NP51 and *P. freudenreichii* NP24 (4×10^9 cfu/animal/day) resulted in a 7.6% increase in average daily milk yield in Holstein cows [306]. A quantitative meta-analysis of 110 papers, 157 experiments and 376 treatments [307]

followed by another meta-analysis [308], revealed that dietary supplementation of *S. cerevisiae* increased milk and fat yield significantly ($P < 0.05$). DM intake was increased by 0.62 kg/day during early lactation and 0.78 kg/day during late lactation. Increased feed intake together with improved microbial digestion of feed could be the possible mode of action for improved animal performance. A combination of *L. acidophilus* NP51 and *P. freudenreichii* NP24 improved the digestibility of CP, NDF and ADF in lactating Holstein cows resulting in increased milk production/day by 7.6% without increase in DMI [306]. No differences in DMI, BW gain; milk yield and milk composition was observed when yeast was added in a diet based on finger millet (*Eleusine coracana*) straw [309] or maize-silage-based diet [310] for lactating crossbred cattle. The absence of a response underlines the need to control the conditions under which these additives are applied, most importantly, diet composition, rumen pH and possibly overall stress [311, 312].

The DFM or their cell wall components have a well-recognized effect on the immune function in monogastric animals. Mannan oligosaccharides have shown to act as a ligand offering competitive binding sites for Gram-negative bacteria allowing removal from the digestive system. β -Glucans have exhibited immune-modulatory effects. Recently, feeding yeast fermentation extracts to transition cows demonstrated that DFM can effectively induce the nonspecific immune system postpartum, but do not seem to affect immunoglobulin concentrations [313]. A number of commercial products based on DFM are indeed being used as immune-stimulants in dairy cow rations.

Monensin/rumensin

It is a common rumen modifier provided either through feed premix or in controlled-release capsule (CRC) form. It acts by interrupting transmembrane movement and intracellular equilibrium of ions in certain classes of bacteria and protozoa that inhabit the gastrointestinal tract. This provides a competitive advantage for certain microbes at the expense of others, altering the rumen microbiota as a result [314]. In modifying the rumen microbial populations, rumensin can enhance the glucose status of dairy cows through increased production of propionate. Consistent with such an effect, administration of monensin by CRC during the transition period and early lactation decreased the incidence of SCK (>1.2 mM) in dairy cows by 50% [315]. Over-conditioned cows (BCS >4.0 at 21 days before expected calving, on a 5-point scale) supplemented with the monensin CRC produced significantly more milk than did unsupplemented controls during early lactation [316]. In a subsequent experiment, cows administered the monensin CRC had decreased circulating concentrations of NEFA during the week immediately preceding calving, but concentrations were not affected during the first week post-calving [317]. In contrast, cows fed 300 mg monensin/day from 28 days before calving until calving did

not alter concentrations of NEFA or glucose during the prepartum period, but decreased circulating NEFA concentrations during the first week post-calving and tended to have increased DMI during early lactation [318]. Mullins et al. [319] reported that monensin supplementation by daily top-dress on TMR (400 mg/day) from -21 through +21 days postpartum decreased both mean and peak concentrations of BHBA and liver TG accumulation. Duffield et al. [320–322] conducted a meta-analysis of the effects of dietary supplementation or ruminal administration of monensin on health and production of dairy cows. Across 59 studies, they determined that monensin supplementation reduced circulating BHBA and NEFA concentrations [320]. Across 77 studies, they determined that monensin supplementation decreased DMI slightly but increased both milk yield and milk production efficiency [321]. Finally, across 16 studies, monensin supplementation decreased risk for both ketosis and left displaced abomasum; administration by CRC resulted in greater reductions in retained placenta and metritis than did administration in the diet [322].

Phased Feeding

Nutritional-management plays a primary role during the transition phase of high-yielding bovines. It can be managed by following the phased feeding programme, which may be either a three-phase (far-off dry cows, close-up dry cows and fresh cows) as developed by NRC [40] or a six-phase (far-off dry cows, close-up dry cows, fresh cows, early lactation cows, mid-lactation cows, late lactation cows) as recommended by Illinois university extension [323] or the one recommended by Israeli Dairy Board [324]. General guidelines for application of phased feeding are given below.

Far-off dry phase

During this period cow's mammary glands dry up and BW improves. TMR should contain 11–12% CP, 1.2–1.3 Mcal NE_L/kg DM, <0.7% calcium, <0.35% phosphorus and <1.5% potassium. DMI should be 1.8–2.5% of BW and average daily gain should not exceed 450 g/day. Higher levels of concentrate may be offered if cows are weak, growing and have not reached mature size, or during environmental stress, and/or when low quality forage is fed. The TMR should contain about 25–30% silage (DM basis) so as to improve the ration palatability.

Close-up dry phase

During this phase level of sodium should be low and the TMR should be supplemented with anionic mineral mixture, 100 g calcium/day, yeast culture (*S. cerevisiae*),

6–8 g niacin/day and 100–150 g calcium propionate starting 3–7 days before calving if SCK occurs.

Fresh cows after parturition

The density of the ration should be increased to check low DMI. TMR should contain UDP and digestible PEF, such as soy hulls or citrus pulp and should be supplemented with protected nutrients, 12 g niacin/day, 1.5% sodium bi-carbonate and magnesium oxide as buffer, yeast culture (*S. cerevisiae*), and calcium propionate for maintaining blood glucose. Frequency of feeding should be at least three to four times per day.

Early-mid lactation cows

TMR should be cereal rich and should contain high-quality forage to improve DMI. It should contain 16–16.7% CP, out of which about 34–36% should be UDP and supplemented with protected nutrients. Maximum individual intake of NDF by high-yielding cows should be 1.3–1.4% of BW. The forage NDF should not constitute more than 17–18% of total DMI, but can be increased to 30–34% by using wheat bran, orange peel or citrus pulp. The BPF content should not exceed 10–15 g/kg DMI. In mid-lactation, the need for feed additives should be reviewed.

Late lactation cows

These cows are pregnant, gaining weight and milk production is declining. The proportion of forages should be increased in the ration. Supplemental sources of UDP, BPF and other feed additives should be eliminated.

Conclusions

Nutritional management plays a pivotal role during the transition phase (3 weeks on either side of the parturition) in high-yielding bovines. Nutritional imbalances, caused by the low quality of desired nutritious rations, reduced appetite and low intake are responsible for depressed performance specifically during the transition period. Other factors such as lack of knowledge and skill of farmers in feeding-management during this phase exacerbate the situation in most of the developing countries. Majority of the farmers boost up the feeding of animals only after parturition. The farmers should be advocated to use proper type of feeding-management required during the transition phase. The approaches discussed in this paper and in particular phased feeding programme would help increase milk production as well as profit for the farmers. These strategies would help overcome hurdles associated with the transition phase, i.e. drop in DMI, abrupt fall in

milk production, loss in BW and body condition score, and metabolic and infectious diseases.

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